

SABHA, Libyan Arab Jamahiriya

WMO: 621240

Lat: 26.987N

Lon: 14.473E

Elev: 435

StdP: 96.21

Time Zone: 2.00 (E02)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	5.9	-10.9	1.6	21.0	-8.6	1.9	20.3	16.9	17.5	14.1	18.7	1.9	100	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.8	42.5	20.1	41.2	20.0	40.2	19.8	22.6	38.0	21.9	37.6	21.2	36.9	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.0	31.0	16.1	12.1	31.0	15.0	11.2	31.0	68.6	38.1	65.8	37.5	63.4	36.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	11.5	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	13.6	15.0	19.8	25.2	29.5	32.2	31.9	31.7	31.0	25.5	19.2	14.2
	DBStd	7.75	2.85	3.68	4.37	3.88	3.88	2.92	2.15	1.74	2.61	3.46	3.07	3.21
	HDD10.0	4	2	1	0	0	0	0	0	0	0	0	0	2
	HDD18.3	457	151	108	38	2	0	0	0	0	0	0	24	135
	CDD10.0	5155	113	142	302	457	605	664	679	673	629	481	276	133
	CDD18.3	2568	4	16	82	209	347	414	421	415	379	223	50	8
	CDH23.3	37393	41	181	942	2571	5123	6627	6697	6509	5626	2517	472	86
	CDH26.7	22969	4	65	395	1351	3216	4452	4375	4180	3531	1240	147	14
Wind	WSAvg	5.0	4.4	4.3	5.3	6.1	6.2	5.8	5.5	5.1	5.2	4.8	4.0	3.7
Precipitation	PrecAvg	8	1	1	1	1	1	0	0	0	0	1	1	1
	PrecMax	50	26	14	16	7	7	15	0	0	6	10	9	18
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	10	4	2	3	2	2	2	0	0	2	2	2	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	32.0	36.0	40.1	43.0	45.2	43.2	42.4	42.4	39.7	33.2	28.4
		MCWB	13.5	15.6	16.5	18.9	19.6	21.2	20.0	20.3	20.4	19.0	16.3	13.0
	2%	DB	24.0	28.5	33.2	37.5	41.0	42.8	41.5	40.7	40.9	36.5	30.7	26.0
		MCWB	11.5	14.3	16.3	17.7	19.4	20.3	20.4	20.3	20.4	18.6	15.6	12.1
	5%	DB	22.1	24.9	30.9	35.7	39.5	41.2	40.1	39.3	39.6	34.4	28.2	23.2
		MCWB	11.3	12.6	14.9	17.4	19.1	19.8	20.4	19.9	19.8	18.0	14.7	11.9
	10%	DB	20.0	22.1	28.6	33.6	37.8	39.9	38.7	38.2	38.2	32.8	26.0	21.1
		MCWB	10.6	10.9	13.8	16.3	18.4	19.4	19.9	19.7	19.6	17.5	14.0	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	16.4	18.1	21.1	22.2	23.1	23.7	23.3	23.3	21.1	19.3	15.3
		MCDB	24.2	30.7	33.2	37.6	39.1	41.4	38.2	37.0	37.3	34.4	28.1	24.9
	2%	WB	13.3	14.6	16.7	19.4	21.0	22.0	22.5	22.1	22.1	20.0	17.2	13.7
		MCDB	22.3	26.7	31.2	34.5	37.8	40.0	37.9	37.0	36.6	32.8	28.4	22.5
	5%	WB	12.1	13.1	15.4	18.1	20.0	21.0	21.7	21.3	21.2	19.1	15.8	12.7
		MCDB	20.3	23.5	29.5	32.5	36.2	39.1	37.0	36.3	36.3	31.5	25.8	21.2
	10%	WB	11.0	11.8	14.3	17.2	19.1	20.1	20.9	20.7	20.4	18.3	14.7	11.8
		MCDB	18.8	21.0	27.1	31.0	34.8	37.7	36.2	35.8	35.8	30.6	24.1	20.1
Mean Daily Temperature Range	5% DB	MDBR	12.4	13.0	14.1	13.9	13.6	13.8	12.9	12.7	13.2	12.7	12.8	12.8
		MCDBR	14.6	15.8	16.8	16.0	14.7	14.9	14.1	13.8	14.4	14.1	14.6	15.5
		MCWBR	7.2	7.4	7.7	6.6	5.7	5.8	5.5	5.1	5.4	5.6	6.4	7.5
	5% WB	MCDBR	13.4	14.7	15.6	15.0	14.2	14.6	12.9	12.7	13.5	13.0	13.1	13.7
		MCWBR	7.6	7.5	7.5	6.8	6.3	6.4	5.8	5.6	6.1	5.6	6.9	7.5
Clear-Sky Solar Irradiance	taub	0.370	0.409	0.497	0.611	0.645	0.587	0.489	0.496	0.536	0.486	0.417	0.374	
	taud	2.084	1.970	1.746	1.549	1.507	1.628	1.851	1.849	1.751	1.846	1.976	2.087	
	Ebn at Noon	859	854	798	721	698	735	810	803	754	765	797	833	
	Edh at Noon	140	170	226	282	295	260	207	206	221	190	155	134	
All-Sky Solar Radiation	RadAvg	4.30	5.16	6.28	6.91	7.36	7.89	8.01	7.45	6.61	5.39	4.47	3.95	
	RadStd	0.22	0.29	0.27	0.26	0.30	0.20	0.08	0.09	0.16	0.21	0.12	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page